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# Targeting in Hamiltonian systems that have mixed regular/chaotic phase spaces

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The problem of directing a trajectory of a chaotic dynamical system to a target has been previously considered, and it has been shown that chaos allows targeting using only small controls. In this paper we consider targeting in a Hamiltonian system, whose phase space contains a mixture of regular quasi-periodic and chaotic regions. A multistep forward–backward method targeting strategic intermediate points is found to be efficient and robust. It takes full advantage of the phase space structure and is believed to yield optimal transport times. It is robust under the influence of small noise and small modeling errors and recovers from temporary loss of control. Two illustrative examples, the standard map and the restricted circular three body problem, are presented. (The latter corresponds to motion of a space probe in the presence of the earth and the moon.) Comparisons are made of our method to other targeting strategies. © *1997 American Institute of Physics.* [S1054-1500(97)01604-2]



**Directing a dynamical system to a desired state as quickly as possible is an important goal in many engineering applications. With the development of nonlinear control techniques, this so-called ''targeting'' has become feasible, using small, inexpensive controls. Applications of nonlinear targeting arise, for example, in space flight, where a spacecraft has to reach its destination quickly, using as little fuel as possible. Due to the particularly complicated dynamics of the (frictionless) mechanics involved, conventional targeting techniques often become infeasible. The targeting technique developed in this article takes full advantage of the dynamical structure of these systems and yields robust, time optimal trajectories to a desired target. As an example, a time and fuel optimized spacecraft trajectory to the moon is calculated.**

## I. INTRODUCTION

In recent years much attention has been paid to the control of chaotic dynamical systems using small perturbations.[1–4] One type of chaos control attempts to enhance the time averaged performance of a system by the stabilization of unstable periodic orbits embedded in the chaotic set. In addition, another type of chaos control has also been used, in which the goal is to direct trajectories to a desired target in the phase space. The latter problem has been called *targeting*[5–10] and will be the subject of this article.

While there has been a strong focus on dissipative systems in the past, Hamiltonian systems present new challenges to chaotic targeting, due to their complicated transport properties. While dissipative systems exhibit ergodic behavior on a strange attractor, ''soft'' chaotic Hamiltonian systems have a complicated phase space structure with coexisting and interwoven chaotic and regular (quasi-periodic) regions. There is no structural stability and the transport times can vary greatly over phase space. In the chaotic regions targeting using small controls is possible, but due to the existence of barrier regions across which there are very long transport times (e.g., the so-called cantori regions), straightforward application of previous methods[5,6] presents problems (see Section II B).

One solution to targeting under these difficulties has been proposed by Bollt and Meiss.[8–10] Their method takes advantage of the fact that long trajectories in a compact phase space are recurrent. Starting with a long unperturbed trajectory that eventually reaches the target, they use small well chosen perturbations to cut recurrent loops from the trajectory, shortening it whenever possible. This method significantly reduces the transport time, although there is little control over its performance, i.e., how close the obtained transfer time comes to the optimum is in practice difficult to determine. In spite of this, in the examples they consider it copes well with the complications arising from Hamiltonian dynamics. Furthermore, their method is applicable in cases where a model of the system is not available and the control strategy is to be learned form observational data.

In this article we will follow another scheme, which we call the *pass targeting method*. In the following, we consider two-dimensional Hamiltonian maps or Hamiltonian flows with two degrees of freedom. Taking explicit advantage of their well-known characteristic phase space features, we find the regions of phase space, through which the orbit must go in order to reach the target, and between which natural transport is typically slow. These regions may be compared to passes in mountainous regions, which have to be traversed by the orbit in order to move between different parts of phase space. The time for a typical trajectory to reach such a pass may be large. However, there are special short orbits that quickly lead through a pass. The pass targeting method uses the passes between source and target as intermediate target

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regions and targets them consecutively, finding each time the shortest trajectory connecting them.

For the two-dimensional Hamiltonian maps considered in what follows, the passes are easily identified with the overlapping parts of certain resonance regions[11,12] that form natural barriers to the transport. In the following we construct a network of these passes connecting each point of the connected chaotic region to another one by successive forward–backward steps between adjacent passes in the network. Having been presented with source and target, we find the fastest connection through the network. We believe that our method typically gives an orbit that is globally optimal or very nearly globally optimal. In particular, following this scheme, the time required to reach the target grows only logarithmically with decreasing size of the maximal control and target region, as compared to a typical scaling of the time as the inverse of the target region size for an uncontrolled trajectory. We also find that our method yields good results even under the influence of noise, modeling errors and in cases of very slow transport. In Ref. 8, E. Bollt considered a similar method, but his version of the method did not work well, and he discarded it in favor of targeting through recurrence. We do not understand why in Ref. 8 unfavorable results were obtained for the method. Our method for selecting the passes for intermediate targeting is different from that used in Ref. 8, and this may perhaps account for the difference.

This article is organized as follows. In Section II we describe the pass targeting method in detail. The forward–backward method is reviewed and refined in Section II A. Section II B describes the general pass targeting method, using the standard map in Section II C as an example. As a second example, we apply the method to the restricted three body problem with an application to spacecraft flight to the moon in Section III. For both examples we compare our results with earlier publications. A general discussion is given in Section IV.

## II. DESCRIPTION OF THE PASS TARGETING METHOD

The control of a dynamical system is usually applied by changing some accessible system parameter or by adding an external force. (In the latter case, the system may be formally regarded as having been altered such that the size of the forces are ''parameters'' of the new system.) In space flight applications, for example, impulsive thrusts of a rocket engine are commonly used for control of the trajectory. With this application in mind we assume, for simplicity's sake, direct access to the phase space momentum variables to achieve control. However, all the following results can also be obtained using other control variables.

In this section we will describe our new targeting procedure and restrict ourselves to targeting problems in the chaotic region of the phase space of a Hamiltonian system. At the heart of the pass targeting method lies the successive piecewise application of the forward–backward method in consecutive small steps. What we mean by small is clarified

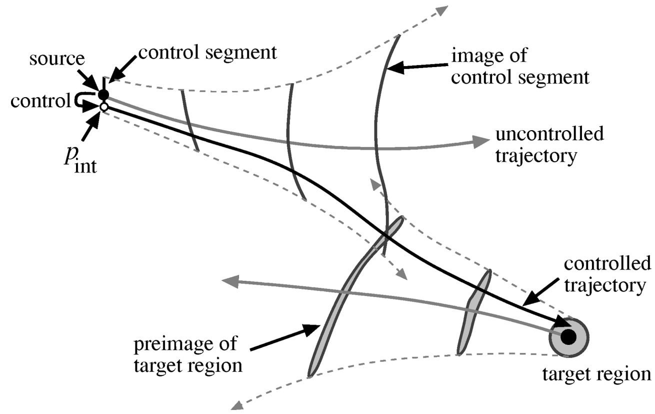


FIG. 1. Forward–backward method: The control segment is evolved forward in time, while the target region is iterated backward, until the image of the control segment intersects the preimage of the target region. The point $p_{\text{int}}$ is the point on the original control segment that leads to the intersection point. Setting the control to $p_{\text{int}}$ directs the trajectory to the target region.

in the following. Longer distances between source and target are subdivided into smaller parts that can each be targeted using a forward–backward step. The subdivision is done by constructing a connected network of passes (regions that have to be traversed by an efficient trajectory), which can be reached by successive forward–backward steps.

### A. Forward–backward method

In this section the forward–backward method introduced by T. Shinbrot *et al.* in Ref. 5 is reviewed. We assume a small phase space curve $(p,q)(s)$ of length $\delta$ centered around the source point $(p_0,q_0)$ to be directly accessible for control. (Here we think of $p$ and $q$ as being momentum and position variables.) In practice the $\delta$ length control segment might simply be a small momentum interval centered around the momentum of the source point. For simplicity, we consider two-dimensional maps only.

In order to find the value of the control that guides the trajectory to a small target region of radius $\varepsilon_t$, we iterate the control segment forward in time. Likewise, the small target region is iterated backward in time. Assuming there exists a trajectory that transits from the source to the target, the two will eventually intersect. Given an intersection, we determine the location $p_{\text{int}}$ on the original control segment that evolves to the intersection point (see Figure 1). Setting the control value to that point $p_{\text{int}}$, the trajectory is directed to the target region. Note that $p_{\text{int}}$ corresponding to the first intersection is the initial condition to the fastest natural trajectory between source and target region that can be accessed by the control.

In previous practice the control segment is iterated until the size of its image has reached the size of the typical phase space extension.[5] Then the target region is iterated backward until its preimage intersects the image of the control segment. The typical size of the image and preimage required to find an intersection varies greatly over Hamiltonian phase space. Therefore we slightly modify the scheme above: Rather than taking all the forward steps at once, we take one forward step and check for an intersection with the target region. If no intersection occurred, we calculate the sizes of both the image of the control segment and the target region. The smaller of the two is then iterated one step forward or backward, respectively. This is repeated until an intersection occurs. This way a nominal size of the image of the control segment need not be chosen in advance.

As the curve segment is evolved forward in time by $n_1$ steps, its length grows roughly exponentially $\sim\delta\exp(\lambda_1 n_1)$, where $\lambda_1$ is the positive pointwise finite-time Lyapunov exponent for forward evolution from the source $(p_0,q_0)$. Likewise the small ball of diameter $\varepsilon_t$ around the target point stretches exponentially $\sim\varepsilon_t\exp(|\lambda_2|n_2)$ in one direction as it is iterated $n_2$ steps backward in time. Here $\lambda_2$ is the negative pointwise finite-time Lyapunov exponent for backward evolution from the target point. (Although the absolute value of the infinite-time positive and negative Lyapunov exponent are identical for Hamiltonian systems, the finite-time Lyapunov exponents $\lambda_1$ and $|\lambda_2|$ may be different.) The transfer time $\tau$ obtained in this way is of the order of

$$\tau\sim\tau_1+\tau_2=\lambda_1^{-1}\ln(L/\delta)+|\lambda_2|^{-1}\ln(L/\varepsilon_t), \qquad (1)$$

where $L$ is the typical length scale on which intersections occur.

In practice the image of the control range is calculated for a discrete number of control settings, in between which it is approximated using linear interpolation. Similarly the preimage of the target region is discretized by iterating evenly spaced points on its perimeter backward in time. When the first intersection is found, its exact location is approximated by bisectional refinement and the control is set to the location $p_{\mathrm{int}}$ on the original $\delta$ segment corresponding to the $n_1$ preimage of the intersection. Applying this control to the system directs the trajectory to the target region in optimal time. We call a trajectory *optimal*, if it is as fast as the fastest natural trajectory between the initial control segment and the target region. Thus forward–backward control is optimal by construction. [Note that this does not preclude the possibility that there may still be controlled orbits that are faster. It appears to us, however, that *with small control* controlled orbits that are more than marginally faster are unlikely if (as is the case in our moon targeting example, Section III) orbits between the chosen passes are robustly chaotic.]

### B. Connecting source and target via passes

Problems with direct forward–backward targeting may occur, when source and target lie in two distant regions separated by slow transport barriers. Then the time before an intersection occurs is very large. Intersection does not occur until the forward and backward images make contact through the slow transport barrier. During that time the extensions of both the image of the control segment and the preimage of the target region grow exponentially, requiring an exponentially increasing number of discrete points to resolve them. This problem can be overcome by successively targeting the passes between resonances that connect the phase space regions of source and target. Since forward–backward target-

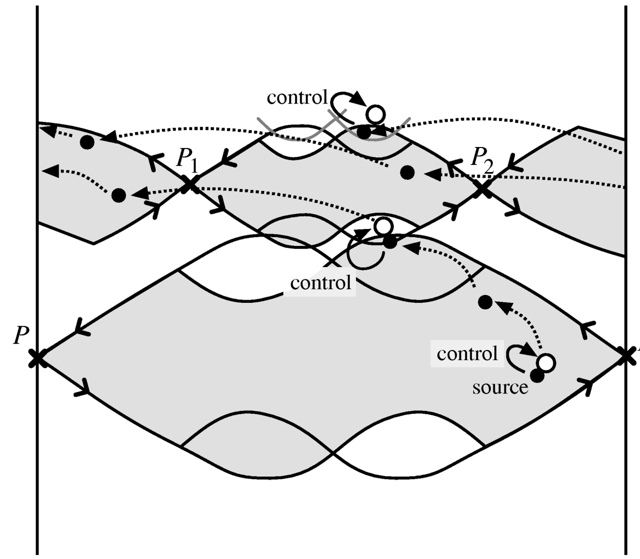


FIG. 2. Associated with each resonance there is an unstable periodic orbit, whose stable and unstable manifolds are used to delineate the boundary of the resonance. To travel from one resonance to the next, the small overlapping region in between has to be traversed. Its vicinity forms a pass. Starting at the source in the lower period one resonance, a small control is applied to target the small overlap region to the upper period two resonance. As this small region is traversed, the overlap region to the next resonance is targeted, applying a second control. There, another control is applied to reach the next target and so on.

ing between the passes is optimal and the passes have to be traversed by the shortest trajectory, the resulting trajectory is globally optimal.

For two-dimensional Hamiltonian maps the natural transport is well-understood,[11,12] and this facilitates the identification of the appropriate passes. In particular, the idea of a ''resonance'' is especially important. This concept is illustrated in Figure 2, where we show a schematic illustration of two overlapping resonances. Each resonance has an associated saddle unstable periodic orbit. For example the period one orbit $P$ is associated with the period one resonance (the lower shaded region in Figure 2), while $P_1$ and $P_2$ form the unstable period two orbit associated with the period two resonance (upper shaded regions in Figure 2). As shown in Figure 2, the stable and unstable manifolds of the periodic orbit delineate the corresponding resonance. To travel from one resonance to the next the small overlapping region must be traversed. Its vicinity forms a pass. A trajectory connecting two distant points might have to cross several resonances, successively traversing passes in between. This is illustrated schematically in Figure 2, where an initial source point in the period one resonance is indicated. Applying a small control the source state is perturbed to the state shown as an open circle in Figure 2, such that the orbit from this state targets the overlap region between the period one and period two resonances. Once the orbit reaches the overlap region a second control is applied to bring it to the next overlap region (shown at the top of the period two resonance in Figure 2) and so on.

Typically, only a few resonances need to be considered as intermediate targets. For example those resonances, whose

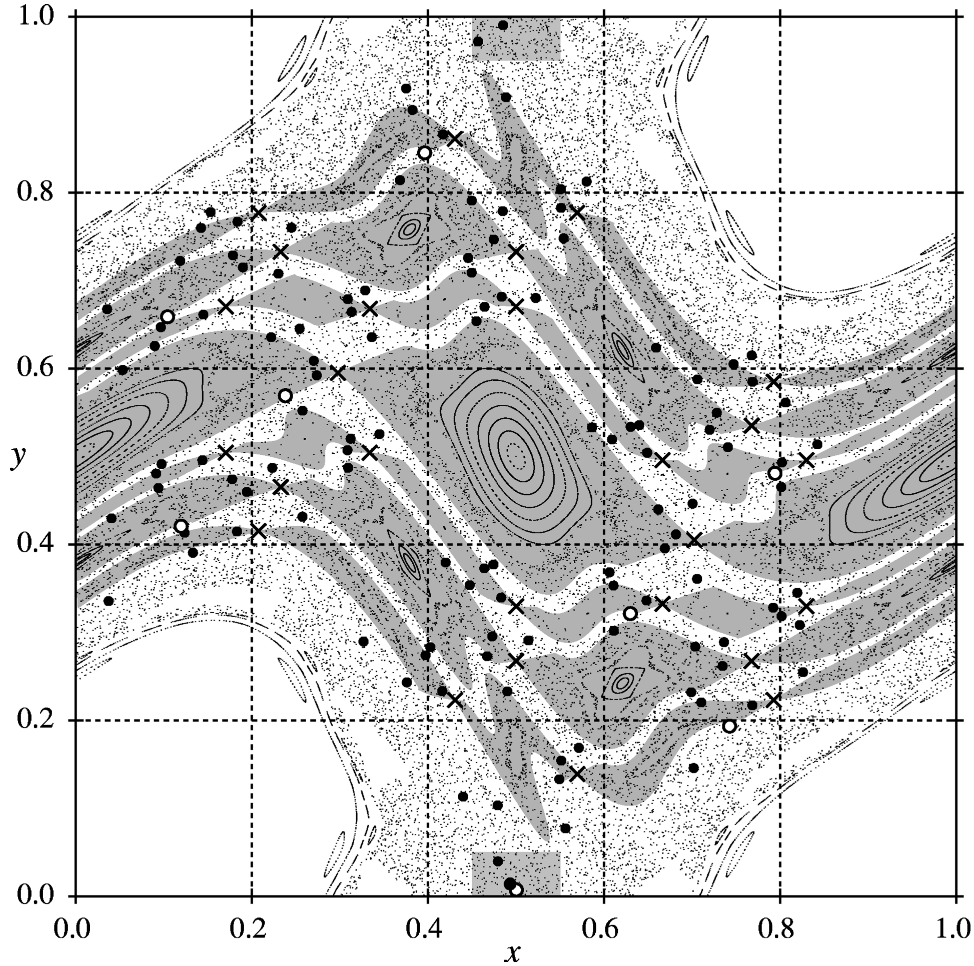


FIG. 3. Phase space diagram of the standard map at $k=1.25$. The seven resonances spanning from left to right that account for slow transport in $y$-direction are shaded in gray. The corresponding unstable periodic orbits are marked as crosses. Starting in the lower gray rectangle, a trajectory is guided to the upper gray rectangle, using 8 successive forward–backward steps: Seven used to traverse the seven resonances in between source and target and one to approach the final target. The solid circles denote the trajectory of length 125 with controls applied at open circles.



neighboring resonances on both sides have no mutual overlap, may be relevant to our method. We call those the *significant* resonances, because the trajectory has to traverse them. In practice, not all of these resonances may be required, and in order to keep the number of intermediate targets at a minimum, all intermediate points that can be skipped are automatically omitted.

### C. Targeting the standard map

Using the standard map as an example, we now illustrate the previous notions and describe the method in more detail. Consider the standard map, Equation (2) with $\delta_n=0$.

$$\begin{pmatrix} y_{n+1} \\ x_{n+1} \end{pmatrix} = \left( \begin{array}{c} y_n - \dfrac{k}{2\pi}\sin(2\pi x_n)+\delta_n \\ \left( y_n - \dfrac{k}{2\pi}\sin(2\pi x_n)+x_n \right) \mathrm{mod} 1 \end{array} \right). \qquad (2)$$

For $k>k_c\approx 0.971\,635$, the standard map has a connected chaotic region that extends over the whole $y$-axis. However, the natural transport in $y$-direction is small, when $k-k_c>0$ is small. Note that (2) with $\delta_n=0$ results from a simple mechanical model [the kicked rotor (e. g., see Ref. 13)], and that for that model $y$ is a momentum. Thus we assume $y$ is accessible for control, as represented by the term $\delta_n$.

To be able to compare our results with those of the recurrence targeting method of Bollt and Meiss,[9] we choose the same parameter settings as in Ref. 9. For $k=1.25$ a periodic cell of phase space is depicted in Figure 3. We consider targeting from a random point in the lower gray rectangle to the upper gray target rectangle, both of which lie in the connected chaotic region. In this example, the natural transport is too slow to allow for computationally efficient forward–backward targeting between source and target directly. In fact, there are seven significant resonances that have to be passed by a trajectory connecting the two regions. These resonances each span from left to right and are depicted as gray regions in Figure 3. There are two period four, two period three and two period five resonances and one period two resonance. The unstable periodic orbits associated with the resonances are shown as crosses, and their stable and unstable manifolds form the boundaries of the resonances (Figure 3). The gray shaded regions in Figure 3 correspond to the gray regions in the schematic diagram in Figure 2.

Following the targeting scheme described in Section II B, we have to find the passes between the seven resonances and use them as intermediate target regions. Finding these passes directly is rather inconvenient. However, since the passes have to be traversed when targeting a point in the neighboring resonance, a much easier procedure is the following. As the target region in every resonance, we choose a small ball around the unstable periodic orbit that is associated with it (depicted as crosses in Figure 3). Any other target region in the chaotic part of that resonance would be equally suitable, but the unstable periodic orbit can be easily found and is guaranteed to lie in the chaotic regime of the resonance. Targeting the unstable periodic orbit from a neighboring resonance forces the controlled trajectory to eventually fall into the pass region (cf. Figure 2). At that point, we abandon the trajectory leading to the unstable periodic orbit and apply a control to direct the trajectory to the next target, i. e., the unstable periodic orbit of the next resonance. This way, the switching takes place at the pass region and optimal transport to the next resonance is guaranteed.

In order to find the iterate, at which the trajectory traverses the pass, we try to target the next resonance at each step, calculating the total time required to reach it from the current source. The switching is performed at the iterate for which this time is minimal.

For each forward–backward step, we choose the control segment in the $y$-direction and allow for a maximal control amplitude of 0.003 (i.e., $|\delta_n| \leqslant 0.003$).[9] By consecutively targeting the passes between two neighboring resonances using the forward–backward technique, the final target is reached, for example, after 125 iterations. The targeted trajectory is shown as solid circles in Figure 3, with controls applied at the positions marked by open circles. Except for the first control each of these points in Figure 3 is located near the lower border of a resonance, i.e., in the pass region at its entrance (see also Figure 2).

For 50 source points chosen at random in the gray source rectangle, the controlled trajectories reached the target rectangle after 125 to 132 iterations, while the uncontrolled transport times varied greatly between 1119 and $3.77\times 10^6$ steps. Transport of less or equal to 126 iterations was achieved in 35 out of 50 cases. The variation in controlled transport time originates form different minimal transport times from the source to the first pass. The rest of the controlled trajectory is nearly unaffected by changes in initial conditions. Since the control range is smaller than the size of the region, from which the initial conditions are picked, different optimal transfer times can arise depending on the exact initial condition. Therefore optimality is suggested even for the slightly longer trajectories of 132 iterations.

For the same parameter settings Bollt and Meiss report an orbit of 131 iterations as their shortest, with others ranging up to lengths of 251.[9] This shortest orbit agrees very well with our results and might be optimal as well, depending on the exact location of the initial condition given.

To test the targeting procedure in a case of extremely slow transport, we also studied transport between the same regions for $k=1.01$. In this case 39 relevant overlapping resonances were found to be responsible for the transport properties. Targeting their unstable periodic orbits as intermediate target points and using 0.005 as the maximal control amplitude (i.e., $|\delta_n| \leqslant 0.005$), a controlled trajectory of 597 iterations was obtained, compared to a trajectory of length 778 achieved by Bollt and Meiss.[9] Variations in the initial conditions affect this result very little, since only the first few steps of the controlled trajectory are affected.

## III. TARGETING IN THE RESTRICTED THREE BODY PROBLEM: FINDING FAST CHAOTIC TRAJECTORIES TO THE MOON

As a more realistic example, we consider targeting in the planar, circular, restricted three body problem[14] to model the dynamics of a spacecraft moving in the earth–moon system. In this model, the mass of the third body (the spacecraft) is assumed to be negligibly small compared to those of the two larger bodies. The two primaries, being affected by their mutual gravitational forces only, are assumed to move on circular orbits about their common center of mass. The restricted three body problem deals with the motion of the third body in the gravitational field of the two primaries.

Although this model is rather crude for actual space flight applications, it is usually used for a preliminary analysis of space missions, and the results obtained are refined later on, using more detailed models (e.g., including the effects of the Sun and other planets). Since we focus on the performance of the targeting method, a refined model is beyond the scope of this article.

Starting at a circular parking orbit near the earth, the classical method to inject a spacecraft into a small-radius circular lunar orbit is the *Hohmann transfer*.[15,16] It consists of two maneuvers requiring large rocket thrusts, one to accelerate the spacecraft to leave the earth and one to decelerate it relative to the moon to achieve lunar capture. One objective of chaotic targeting in this context is to reduce the overall fuel consumption for a transfer to the moon, allowing for a larger pay load and thus reducing the transport cost.

This problem has for example been previously addressed by Belbruno and Miller (see Ref. 17). Making only the first maneuver of the Hohmann transfer and using a lunar swing-by to reach the $L_1$-point of the earth–sun system, they were able to achieve ballistic capture by the moon, sparing the second maneuver of the Hohmann transfer. [The Lagrange points $L_1$ to $L_5$ are the equilibrium states of the restricted three body problem. The $L_1$-point, in particular, is located on the axis between the two bodies, where the centrifugal force together with both gravitational forces on the third body cancel.] This concept was used to save the otherwise failed Japanese Hiten mission, saving about 18% of the thrust required for the Hohmann transfer. This saving was achieved at the expense of a longer transfer time. While the Hohmann transfer takes about three to four days, the Hiten mission reached the moon after 139 days.

Another approach has been pursued by Bollt and Meiss

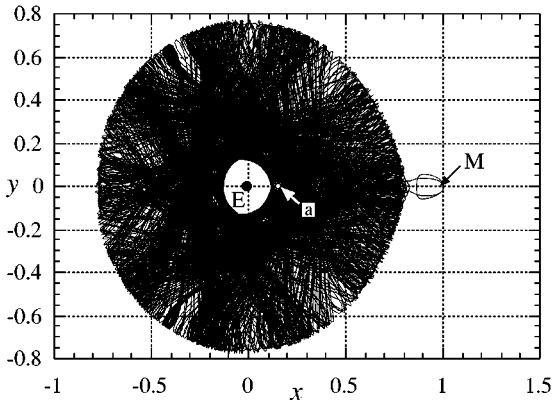


FIG. 4. Starting from a circular parking orbit around the earth, a parallel thrust is applied to inject the spacecraft into a chaotic region of phase space at the point $a$ marked by the open circle. The spacecraft is then left to move freely in the chaotic region, following the depicted trajectory. Because the chaotic region extends to the vicinity of the moon, the uncontrolled trajectory can eventually reach it. In this particular example, it reaches the vicinity of the moon for the first time after approximately 27 years, but rather than approaching a stable orbit around the moon, it quickly leaves the moon again. The trajectory has to be followed for a much longer time (not depicted) until a stable orbit around the moon is approached.

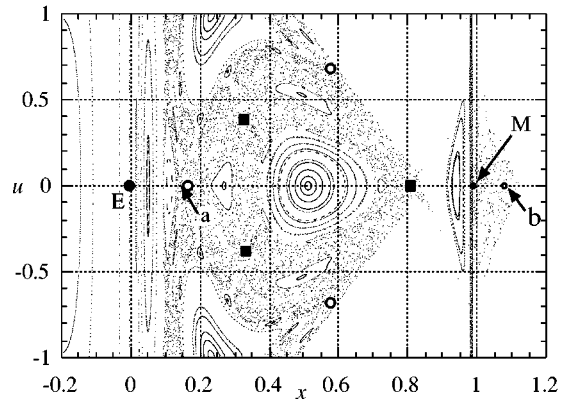


FIG. 5. Poincaré surface of the planar restricted three body problem at $J=-3.179\,48$. The source and the target state are denoted by $a$ and $b$, respectively. The unstable period three orbits used as intermediate target points are denoted by open circles and solid rectangles, respectively.

using chaotic targeting through recurrence.[10] Starting from a circular parking orbit around the earth, they apply a first thrust to put the spacecraft into a chaotic phase space region that extends to the vicinity of the moon. In the chaotic region the spacecraft would follow a trajectory that would eventually approach a stable orbit around the moon. Without control, however, this transfer would be prohibitively long. Figure 4, for example, depicts the first part of such an *uncontrolled* chaotic trajectory. It reaches the vicinity of the moon for the first time after approximately 27 years, but rather than approaching a stable orbit around the moon, it quickly leaves the moon again. The trajectory has to be followed much further, before it eventually approaches a stable orbit around the moon. By removing recurrent loops from such a long trajectory using small control thrusts, Bollt and Meiss reduced the transfer time drastically, while saving about 40% of the thrust compared to a Hohmann transfer from the *same* parking orbit. The transfer time obtained, however, is still very long: 2.05 years.[10]

Using this latter mission plan, we apply our pass targeting method to optimize the time for the chaotic transfer. We use standard coordinates for the restricted three body problem, setting the sum of the masses of the two primaries as well as Newton's gravitational constant to unity ($M=m_{\mathrm{e}}+m_{\mathrm{m}}=1,\ G=1$). In the resulting units, the earth–moon distance is also unity. The mass ratio for the moon and the earth is set to $m_{\mathrm{m}}/m_{\mathrm{e}}=0.0123$.[10] In a frame rotating with the primaries (origin at the center of mass), we fix the position of the moon and earth at $x_{\mathrm{m}}=m_{\mathrm{e}}$ and $x_{\mathrm{e}}=-m_{\mathrm{m}}$ respectively (see Figure 6). In these units, the time and velocity are measured in units of 104 hours and 1024 m/s respectively.

In this frame the equations of motion for the spacecraft are (see for example Ref. 14)

$$\dot{x}=u, \quad \dot{y}=v, \tag{3}$$

$$\dot{u}=x+2v-m_{\mathrm{e}}\frac{x-x_{\mathrm{e}}}{r_{\mathrm{e}}^3}-m_{\mathrm{m}}\frac{x-x_{\mathrm{m}}}{r_{\mathrm{m}}^3}, \tag{4}$$

$$\dot{v}=x-2u-\left(\frac{m_{\mathrm{e}}}{r_{\mathrm{e}}^3}+\frac{m_{\mathrm{m}}}{r_{\mathrm{m}}^3}\right)y, \tag{5}$$

where $(x,y)$ are spatial coordinates in the rotating frame, and $(u,v)$ are the corresponding momenta. $r_{\mathrm{m}}$ and $r_{\mathrm{e}}$ denote the distances of the spacecraft from the moon and earth, respectively. The quantity

$$J=u^2+v^2-(x^2+y^2)-2\left(\frac{m_{\mathrm{e}}}{r_e}+\frac{m_{\mathrm{m}}}{r_{\mathrm{m}}}\right) \tag{6}$$

is a constant of the motion, confining the Hamiltonian flow to a three-dimensional submanifold $J=\mathrm{const}$. To obtain a two-dimensional map from this effectively three-dimensional flow, we construct a Poincaré return map for fixed $J$, mapping the $(x,u)$-plane to itself, whenever the trajectory traverses the $x$-axis with positive $v$. Figure 5 shows such a surface of section.

Starting at a nearly circular orbit of radius 59 669 km ($J=-7.1738$) around the earth, we perform the first parallel impulsive thrust to inject the spacecraft into the chaotic region at $J=-3.179\,48$. [This value for $J$ is chosen slightly above the minimum value $J_{L_1}=-3.1883$, for which a transfer to the moon is possible, but below the value, for which trajectories may escape from the earth–moon system.] This requires a change in velocity of $\Delta V=744.4$ m/s. The spacecraft is now in the state $a$ in Figure 5 amidst the chaotic region.

In a first step we analyze the phase space structure, noting that long uncontrolled trajectories leading to the moon just before accessing the region close to the moon always come close to the inner unstable period three orbit depicted by solid squares in Figure 5. This period three orbit is chosen as one intermediate target point. But before reaching this period three orbit from the source, another resonance corresponding to the period three orbit depicted by the open circles in Figure 5 must be passed. Now, we successively

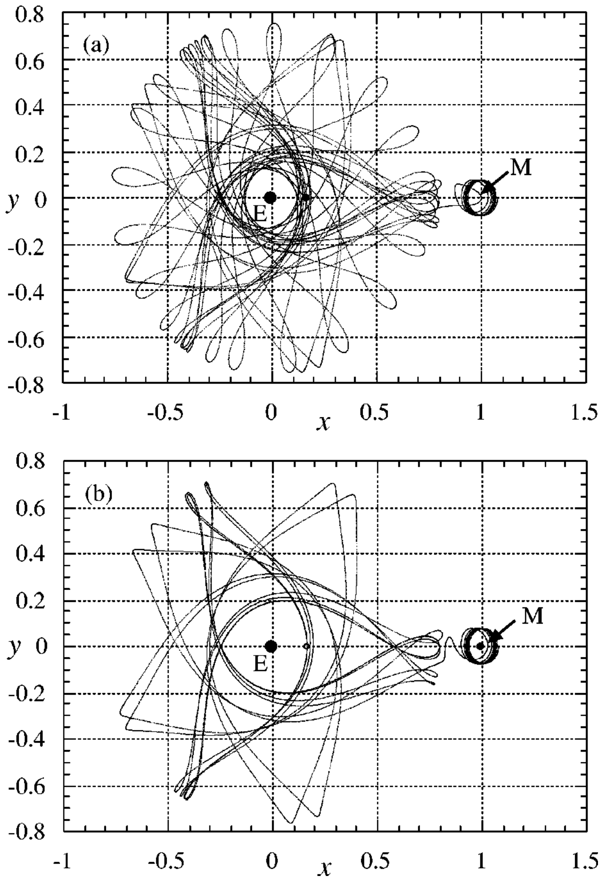


FIG. 6. Controlled chaotic trajectories to a stable lunar orbit. (a) Starting from a circular orbit 59 669 km above the earth's center (time required for transfer: 377.5 days). (b) Starting from the period three orbit denoted by open circles in Figure 5 (time required for transfer: 293 days).

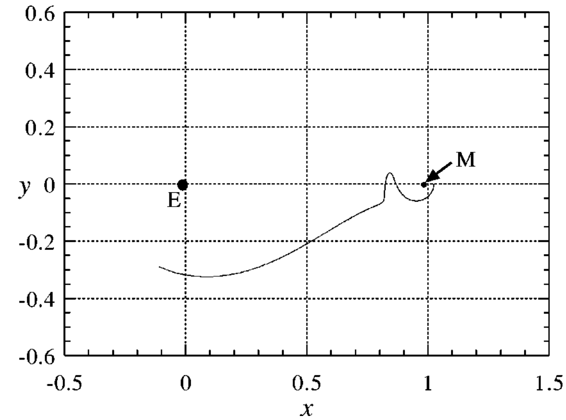


FIG. 7. Part of a chaotic trajectory obtained by pass targeting. The similarity to the orbit used by the LGAS mission (Ref. 17) suggests the applicability of chaotic targeting to electrically propelled spacecraft missions. The time required to follow the trajectory depicted is about 16 days. In the case of the LGAS mission, the starting point of such a trajectory was reached by slowly spiraling outward from a near earth parking orbit using a small, continuous thrust. Likewise, the spacecraft approached the moon by spiraling towards the moon after the transfer.

target the two period three resonances before aiming at the stable elliptical orbit depicted as $b$. Note that in order to keep the spacecraft at constant $J$, the control segment for the forward–backward control is implicitly given by (6). Using the resulting trajectory, the spacecraft reaches the target after only 87.12 time units, corresponding to 377.5 days $\sim 1.03$ years, which is $\sim 50\%$ of the transfer time required in Ref. 10. The trajectory is depicted in Figure 6(a). The total thrust required for controlling the trajectory is $\Delta V_{\text{control}} = 2.48 \cdot 10^{-3} = 2.54$ m/s, and a final stabilizing thrust of $\Delta V_{\text{stab.}} = 2.0 \cdot 10^{-3} = 2.05$ m/s is required to move the spacecraft out of the chaotic regime onto the stable orbit located at $b$. The overall thrust requirement for this transfer is $\Delta V = 748.9$ m/s, which is slightly below the 749.6 m/s required by the transfer in Ref. 10. Thus, within the $\Delta V$-limits given in Ref. 10, the target was reached in about half the time.

Above, we chose the source point for the trajectory rather arbitrarily. In fact, it was obtained in Ref. 10 by ''trial and error,'' and a better source point can be found in its vicinity: Starting on the period three orbit depicted by open circles in Figure 5 results in an even shorter transfer time of 293 days $\sim 0.80$ years, requiring a controlling $\Delta V_{\text{control}} = 0.62$ m/s and $\Delta V_{\text{stab.}} = 2.61$ m/s. The corresponding transfer orbit is shown in Figure 6(b).

The reason this source point yields such a good result is the following: In order to achieve ballistic capture by the moon, the spacecraft must acquire additional angular momentum in order to follow the moon on its circular orbit. At the source point, the angular momentum of the spacecraft is much smaller. The only way the spacecraft can acquire this additional angular momentum is by using the gravitational forces of the moon. The most effective way to do this is to spend as much time as possible close to it. Starting on the period three orbit the trajectory obtained comes close to the moon every period.

The practical use of the trajectories obtained for missions to the moon using chemical propulsion systems, however, should not be overestimated, since the starting distance from the earth must be unusually large in order to take advantage of the chaotic motion. Most of the gain in $\Delta V$ must be spent on reaching the initial source point on the far out circular orbit. Electrically propelled spacecrafts, such as the LGAS mission to the moon,[17] however, slowly gain altitude by spiraling out from the earth, using a small continuous thrust. These missions naturally reach the chaotic region of phase space, where chaotic targeting becomes applicable. In fact, the part of our trajectory, that passes over to the moon depicted in Figure 7, is similar to the corresponding part of the LGAS' orbit.[17] Thus chaotic targeting as discussed in this paper should be very relevant in such situations.

## IV. DISCUSSION AND CONCLUDING REMARKS

While in the example of the standard map the source and target both lie well inside the chaotic region, the target orbit around the moon lies on a stable island. In order to approach such a stable KAM-region, the *uncontrolled* trajectory must pass an infinite hierarchy of smaller and smaller resonances, never reaching the outmost stable KAM-surface.[11] A stable region can therefore never be reached by an uncontrolled trajectory. When approaching a stable KAM-island, the controlled trajectory must basically follow this same dynamical structure from pass to pass, until the resonance regions are smaller than the maximal control. At that scale, the target can be reached immediately by applying an appropriate control and the complicated dynamics in the vicinity of the stable region can be bypassed. In the space flight example of Section III, the resonances surrounding the target stable lunar orbit were very small. Therefore, by applying a small control that was large compared to the extensions of the resonance structure, it was possible to jump onto the stable island directly from the surrounding chaotic region.

Another consequence of the finite size of the control is an effective coarse graining of phase space, which allows a controlled orbit to traverse thin natural barriers of the motion, connecting regions that are mutually inaccessible by the natural flow. For example, by jumping across a KAM-torus (such as the last invariant curve of the standard map for $k$ slightly below $k_c \approx 0.971\,635$, Ref. 18) separated chaotic regions can be connected.

The phase space structure of two-dimensional Hamiltonian maps has the particularity, that any two resonances are uniquely connected through a number of overlapping resonances.[11] Therefore, given a source and a target, the resonances to be traversed and their order is fully determined. This defines a unique targeting path on the network of intermediate target points.

For practical applications of a control method, its robustness against noise and modeling errors is important. We have tested the pass targeting method under various circumstances, including the influence of small noise, modeling errors and the temporary failure of control. If, for example, the trajectory is lost due to the failure of the control, i.e., a temporary malfunction of a rocket engine, the targeting can be easily resumed when the control is again operational. With an intermediate target point in each relevant resonance, the network of intermediate targets can always be accessed from the current state, which is then used as a new source point.

We studied targeting the standard map ($k = 1.25$) under the influence of small additive noise with amplitudes ranging up to $10^{-2} \times \delta$ at each iteration. Without any change to our targeting scheme, all trajectories reached the target. The targeting times obtained were in general not significantly changed, although occasionally a few iterations more were needed to reach the target. (The longest trajectory recorded was 154 iterations long.)

For larger noise amplitudes control could still be achieved by correcting the trajectory as it deviates from the predicted trajectory, due to the influence of noise. These corrections can be applied by repeating the forward–backward procedure after every few steps along the trajectory. In the absence of noise, this procedure does not change the trajectory and no additional controls are applied. In the presence of noise, however, it suppresses large deviations from the nominal trajectory. This scheme is also practical, when the resolution of the control is coarse or in the presence of significant modeling errors.

The influence of modeling errors was investigated for the standard map, using a standard map at $k = 1.25$ to model the ''real'' system at $k = 1.249$. Using the retargeting scheme just described, and retargeting the trajectory every four steps, all trajectories reached the target. Using 50 randomly chosen source points in the gray source region (Figure 3), the transfer times ranged from 126 to 216 iterations.

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